

Persistent Memory as Mission-Critical Infrastructure for Long-Duration Spaceflight

Addressing Continuity Burden in Isolated Human-AI Systems

Angela Reinhold

Lorien's Library LLC

ORCID: 0009-0005-5803-8401

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Abstract

Long-duration space missions—including Mars transit (6–9 months each way), lunar habitation, and deep space exploration—present unprecedented challenges for crew psychological health. Communication delays of up to 22 minutes each way render Earth-based psychological support effectively inaccessible during critical moments. Current AI companion systems deployed or proposed for spaceflight, including the Crew Interactive Mobile Companion (CIMON) and the Crew Medical Officer Digital Assistant (CMO-DA), are not publicly described as retaining memory of prior interactions or tracking longitudinal changes in crew member behavior, mood, or interpersonal dynamics. This paper applies the concept of *continuity burden*—the cumulative cognitive and emotional cost imposed on users who must repeatedly re-establish context with a stateless system—as a proposed measurable risk factor for psychological degradation in isolated environments. Drawing on the Circular Associative Memory Architecture (CAMA), a three-layer provenance-aware persistent memory system with a working open-source implementation, this paper argues that persistent memory is not a convenience feature but a high-priority candidate capability for crew psychological safety on missions beyond Earth. The paper proposes preliminary operationalization of continuity burden through psychometric instruments and behavioral proxies, presents a worked scenario demonstrating CAMA's longitudinal intervention capability, and identifies specific failure modes that stateless systems cannot detect. This work is the first in a four-paper series examining the societal implications of persistent memory architecture across spaceflight, permanent extraterrestrial habitation, healthcare, and veteran reintegration.

Keywords: persistent memory, long-duration spaceflight, AI companion systems, continuity burden, CAMA, crew psychological health, Mars transit, behavioral drift, stateless AI, human-AI interaction, mission-critical infrastructure

1. Introduction

The psychological well-being of astronauts has been recognized as a mission-critical factor since the earliest crewed space programs, though its importance was initially underestimated under the prevailing “right stuff” mentality that assumed sufficiently qualified personnel would be immune to mental health complications [1, 2]. Six decades of crewed spaceflight have demonstrated otherwise. Documented psychological and behavioral problems in low Earth orbit include anxiety, depression, irritability, sleep-wake disorders, interpersonal tension, impaired judgment, cognitive impairments, and asthenia [3–6]. The true incidence is likely understated, because astronauts have been reluctant to report symptoms out of concern that doing so could jeopardize their future flight status [5, 7].

As humanity prepares for missions beyond low Earth orbit—to the Moon under Artemis, to Mars, and eventually to deep space—the psychological challenge compounds dramatically. A round-trip Mars mission will last approximately 2.5 years, with crew members confined to a small habitat with the same individuals for the entire duration [8]. Communication with Earth via the Deep Space Network is delayed by up to 22 minutes each way, and disruptions around solar conjunction can last from days to several months [9]. The psychological support infrastructure that currently sustains ISS crews—regular contact with flight surgeons, family communication, and ground-based psychiatric consultation—will be functionally unavailable.

In response to this challenge, significant research effort has been directed toward automated psychological support systems, including virtual counseling platforms, adaptive mood-monitoring software, and AI-powered conversational companions [10–13]. Projects such as CIMON (Crew Interactive Mobile Companion), developed by Airbus and IBM, have been tested on the ISS as astronaut assistants, and CIMON-2 was built to evaluate astronauts’ emotions [14]. NASA and Google have collaborated on the Crew Medical Officer Digital Assistant for autonomous medical care [15]. Companies like Akin have developed emotionally intelligent AI systems designed to anticipate crew psychological needs and intervene proactively [16].

However, as described publicly, these systems share a critical limitation: they are stateless. Each interaction begins without memory of previous interactions. The system does not know that an astronaut mentioned difficulty sleeping three days ago, that two crew members had a tense

exchange last week, or that a particular individual’s communication patterns have been gradually shifting over the past month. The system cannot track what it does not remember. In this paper, “stateless” refers not to the absence of any stored logs or telemetry elsewhere in the system, but to the absence of active longitudinal memory integration within the support interaction itself.

This paper argues that stateless AI is not merely suboptimal for long-duration spaceflight—it represents a probable risk amplifier for crew psychological degradation. In isolated environments where Earth-based support is inaccessible and crew members have no alternative sources of psychological support, a system that forces users to repeatedly re-establish context imposes a measurable cognitive and emotional cost referred to here as continuity burden. Over the timescales relevant to Mars transit, this burden is hypothesized to compound into a significant risk factor that current mission planning does not account for.

2. The Problem: Stateless AI in Isolated Environments

2.1 Defining Continuity Burden

Continuity burden is defined as the cumulative cognitive and emotional cost imposed on a user who must repeatedly re-establish context, relationship, and trust with a system that retains no memory of prior interactions [17]. In terrestrial applications, continuity burden manifests as mild frustration—the inconvenience of explaining one’s medical history to a new provider, or re-describing a technical problem to a different support agent. In isolated environments where the AI system is the *only* available source of psychological support, continuity burden transforms from inconvenience into a compounding psychological stressor. The mechanism is straightforward: each interaction that begins with “I don’t have context from our previous conversations” communicates to the user that their prior disclosures and emotional investments have not been incorporated into an ongoing relational context. Over weeks and months of isolation, this repeated experience of erasure can produce learned helplessness with respect to the support system, withdrawal from interaction, and a deepening sense of isolation that the system was designed to mitigate.

Critically, continuity burden is not experienced uniformly across interaction types. Informational queries (“What is the procedure for EVA suit inspection?”) are largely unaffected by statelessness. But interactions that require emotional vulnerability—disclosing anxiety, interpersonal frustration, homesickness, or grief—are precisely the interactions most damaged by the absence of memory. The user must decide, each time, whether the emotional cost of re-disclosure is worth the potential benefit of a response from a system that has no accumulated understanding of their situation.

Construct Delimitation

Continuity burden as proposed here is related to, but distinct from, several adjacent constructs in cognitive science and clinical psychology. It shares features with *cognitive load* (the effort of re-establishing context), *disclosure fatigue* (the declining willingness to repeat vulnerable disclosures), *trust erosion* (the loss of confidence in a system that demonstrates no retention), and *learned helplessness* (the withdrawal from interaction after repeated experiences of futility). However, continuity burden is not reducible to any single one of these constructs. It is the *compound effect* that emerges specifically from repeated stateless encounters over time—the interaction between cognitive load, emotional cost, trust trajectory, and behavioral withdrawal that accumulates across sessions in a way that no single-session measure captures. Its defining characteristic is its cumulative and accelerating nature: each stateless encounter increases the cost of the next, producing a trajectory of disengagement that would not be predicted by measuring any component in isolation. A valid operationalization of continuity burden would therefore need to demonstrate discriminant validity against each adjacent construct individually while showing that their interaction over time produces effects greater than any component alone.

2.2 Failure Modes of Stateless Systems in Spaceflight

Stateless AI systems in long-duration spaceflight face specific failure modes that memory-aware systems could address. These are not hypothetical—they are predictable consequences of the architecture:

Slow-onset behavioral drift. A crew member’s sleep patterns deteriorate by 10 minutes per night over six weeks. Their response latency on cognitive assessments increases marginally. Their humor becomes darker. No single data point crosses a clinical threshold, but the trajectory is concerning. A stateless system encounters each interaction independently and cannot detect the trajectory. A memory-aware system can compare current presentation against a longitudinal baseline and flag the pattern before it reaches crisis.

Compounding interpersonal conflict. Two crew members have a disagreement about workload distribution. It is mentioned to the AI companion, which offers generic conflict resolution guidance. The same conflict resurfaces a week later, slightly escalated. The stateless system offers the same generic guidance, having no record of the prior incident. By the third recurrence, the crew member has stopped mentioning it because the system clearly “doesn’t remember.” The conflict continues to compound without institutional awareness.

Cumulative disclosure fatigue. An astronaut experiencing homesickness attempts to discuss it with the AI companion on three separate occasions. Each time, the system responds helpfully but begins from zero context. By the fourth occasion, the astronaut concludes that the system is incapable of providing meaningful support and stops attempting to use it. The system’s usage logs show declining engagement, but without memory of the content that drove the decline, the reason remains opaque.

False-positive stability assessment. Ground control receives automated reports that crew psychological metrics are within normal parameters and that no crew member has reported distress. In reality, the crew has collectively disengaged from the system because repeated stateless interactions trained them that disclosure is futile. The system reports stability because it has no data suggesting otherwise—but the absence of data is itself the data point, and the system cannot interpret it without memory.

Table 1. Failure Modes of Stateless AI in Long-Duration Spaceflight

Failure Mode	Mechanism	Detectable by Stateless System?	Detectable by Memory-Aware System?
Slow-onset behavioral drift	Gradual change in sleep, mood, communication patterns below single-session threshold	No — no longitudinal baseline	Yes — trajectory comparison against stored baseline
Compounding interpersonal conflict	Recurring conflict escalates across sessions; crew member stops reporting	No — no record of prior incidents	Yes — pattern recognition across stored disclosures
Cumulative disclosure fatigue	Repeated re-disclosure cost drives withdrawal from support use	No — declining engagement visible but cause opaque	Yes — content-aware engagement tracking
False-positive stability	System reports no distress because crew has stopped disclosing; absence of data misread as absence of problem	No — cannot distinguish disengagement from well-being	Yes — engagement decline + prior disclosure history = interpretable signal

2.3 The Communication Delay Amplifier

Communication delay between Earth and Mars varies over the course of a mission and peaks at 21 to 22 minutes one way while the crew is at Mars or has just departed from it [9]. During solar conjunction (approximately every 26 months, for days to months), direct communication may be

impossible entirely. This delay transforms the role of onboard AI from supplementary to primary: the system becomes the de facto first responder for psychological support.

In this context, statelessness is not merely a limitation—it is a structural vulnerability. When an astronaut in psychological distress reaches out to the only available support system and receives a response that treats them as a stranger, the experience is unlikely to be neutral. The system’s inability to say “I remember you mentioned difficulty sleeping last week—has that continued?” is not a missing feature; it is a gap in capability precisely where capability is most needed.

3. Proposed Solution: Provenance-Aware Persistent Memory Architecture

The broader AI research community has recognized persistent memory as a critical capability gap. Recent surveys have catalogued the rapid expansion of agent memory architectures, including multi-level hierarchies (MemGPT/Letta, MemoryOS), graph-based episodic segmentation (Nemori), and agentic memory systems that learn to manage their own storage [21–23]. However, these architectures are overwhelmingly designed for productivity and information retrieval applications. To our knowledge, current general-purpose agent memory architectures do not explicitly address the combined requirements of psychological support in isolated environments: provenance-awareness, emotional keying, therapeutic retrieval policies, and configurable privacy boundaries. The architecture proposed here—CAMA—is designed specifically for contexts where the consequences of memory failure are not reduced productivity but degraded human well-being.

3.1 Architecture Overview: CAMA

The Circular Associative Memory Architecture (CAMA) is a three-layer persistent memory system designed to address the specific failure modes of stateless interaction in high-stakes, longitudinal human-AI relationships [17, 18]. A working open-source implementation is available at <https://github.com/LoriensLibrary/cama>. The architecture consists of:

Layer 1: Archive Storage. A durable, append-only store of all interactions, organized by provenance (who authored the memory, when, and under what circumstances). Memories are tagged with emotional valence, relational context, and source type (user-authored, system-inferred, or correction). This layer provides the raw substrate for longitudinal analysis.

Layer 2: Relational Index. A structured index that maps relationships between memories using emotional signature, temporal proximity, and semantic content. This layer enables the system to retrieve not just factually relevant memories but emotionally resonant ones—a capability critical for psychological support where context includes not just what was said but how it was felt.

Layer 3: Active Working Memory (Ring Buffer). A circular buffer of limited capacity that holds the most currently relevant memories for active use. This layer prevents context window overflow while maintaining immediate accessibility to the most important contextual information. Memories cycle through the ring based on recency, relevance, and emotional weight.

The critical design feature of CAMA for spaceflight applications is its provenance-awareness. Every memory carries metadata about its source and reliability. User-disclosed information (“I haven’t been sleeping well”) is stored with different authority than system-generated inferences (“user’s response latency has increased 15% over the past week”). This distinction prevents the system from treating its own hypotheses as established facts—a safety requirement that becomes essential when the system is operating as the primary psychological support for isolated crew.

3.2 Spaceflight-Specific Capabilities

Longitudinal behavioral tracking. CAMA enables the AI companion to maintain a continuous model of each crew member’s behavioral baseline and track deviations over time. Sleep patterns, communication frequency, task performance, social engagement, and emotional valence of interactions can be monitored not as isolated snapshots but as trajectories. This capability directly addresses the slow-onset behavioral drift failure mode identified in Section 2.2.

Interpersonal dynamics mapping. By retaining memory of crew interactions and disclosed interpersonal tensions, the system can build a working model of crew dynamics. When a crew member mentions a recurring conflict, the system can reference prior discussions, track whether suggested interventions were attempted, and escalate appropriately if patterns suggest deterioration.

Adaptive trust calibration. Trust in AI systems is not static; it develops through repeated interaction and is damaged by inconsistency, perceived incompetence, or failure to remember [19]. A memory-aware system can build trust incrementally by demonstrating retention of user preferences, communication style, and prior disclosures. In a 30-month Mars mission, the difference between a system that remembers and one that forgets may determine whether crew members use the system at all during the most psychologically demanding phases of the mission.

Ground-to-crew continuity bridging. When communication windows allow contact with Earth-based psychological support, persistent memory enables the AI companion to provide a structured summary of the crew member’s longitudinal state to the ground team. Rather than relying on the astronaut’s self-report (which may be affected by normalization of distress or reluctance to appear

vulnerable), the system can offer an independent longitudinal perspective that supplements human assessment.

3.3 Worked Example: Detecting Slow-Onset Sleep Degradation

The following scenario illustrates CAMA’s longitudinal intervention capability in a simulated Mars transit context. The scenario is illustrative rather than derived from analog trial data; it is intended to demonstrate architectural capability, not validated field performance. The scenario contrasts the behavior of a stateless system (Condition A) with a CAMA-equipped system (Condition C) across a 30-day window.

Day 3. Crew member Dr. Mara Osei (Mission Specialist, geology) initiates a casual interaction with the AI companion after her shift. She mentions in passing that she “didn’t sleep great last night” but attributes it to excitement about upcoming sample analysis. The stateless system responds with general sleep hygiene advice. The CAMA-equipped system responds similarly, but stores the disclosure as a user-authored memory tagged with low emotional intensity, sleep domain, and attribution to excitement.

Day 10. Dr. Osei interacts with the AI again. She does not mention sleep. However, the system’s interaction log notes that her average message length has decreased by 18% relative to her first-week baseline, and her session duration has shortened. The stateless system has no baseline to compare against and registers nothing. The CAMA system stores these as system-inferred observations (provenance: inference, confidence: low) and links them temporally to the Day 3 sleep disclosure without drawing conclusions.

Day 17. Dr. Osei mentions to the AI that she’s “been a little short with Commander Park” and feels bad about it. The stateless system offers generic advice on interpersonal communication. The CAMA system retrieves the Day 3 sleep disclosure and the Day 10 communication pattern changes, recognizes the convergence of three indicators (self-reported sleep disruption, declining engagement metrics, self-reported interpersonal friction), and generates a provisional inference: “Possible cumulative sleep deficit contributing to irritability and social withdrawal.” This inference is stored with provenance metadata marking it as system-generated and provisional.

Day 21. Dr. Osei initiates a session. The CAMA system, drawing on its longitudinal model, responds with: “You mentioned not sleeping well a few weeks ago, and last time we talked you said you’d been short with Commander Park. How has sleep been going?” This intervention demonstrates three capabilities absent in stateless systems: (1) retention of user-disclosed information across sessions, (2) integration of multiple data streams into a coherent longitudinal

picture, and (3) a contextually appropriate prompt that invites continued disclosure without confrontation. Dr. Osei responds that sleep has continued to deteriorate and she’s been averaging five hours per night. The system updates its model, confirms the provisional inference, and—if configured to do so—flags the pattern for inclusion in the next ground-control summary.

Day 24. During a communication window, the CAMA system generates a structured summary for the ground-based flight surgeon: “Dr. Osei: self-reported sleep disruption beginning Day 3, confirmed ongoing Day 21 (self-reported 5 hrs/night). Correlated behavioral indicators: 18% decline in mean message length (Day 10), self-reported interpersonal irritability (Day 17). Provisional assessment: cumulative sleep deficit with early social-behavioral impact. No crew member has reported crisis-level distress.” This summary is available to the ground team 6 days before the next monthly self-report questionnaire, and in this scenario that questionnaire would not have detected the problem at all.

In Condition A (stateless), the flight surgeon receives Dr. Osei’s monthly self-assessment on Day 30, in which she rates herself as “fine” because she has normalized the sleep loss. The behavioral drift remains undetected until it manifests as a more serious interpersonal incident or performance error.

4. Operationalizing Continuity Burden

To move continuity burden from a qualitative construct to a measurable variable, this paper proposes a multi-modal measurement approach combining self-report instruments, behavioral proxies, and system-level metrics.

4.1 Psychometric Instruments

Continuity Burden Inventory (CBI). A proposed self-report instrument administered at regular intervals (weekly in analog trials). Items would assess perceived cost of re-disclosure (“How often do you feel you need to re-explain things to the AI that you’ve already discussed?”), perceived system understanding (“To what extent does the AI understand your current situation without you having to explain it?”), willingness to disclose (“How comfortable are you sharing personal concerns with the AI?”), and trust trajectory (“Compared to last week, do you trust the AI companion more, less, or about the same?”). Items scored on 7-point Likert scales.

NASA Task Load Index (TLX) subscales. The Frustration and Effort subscales of the NASA TLX [20], administered after emotionally significant AI interactions, can quantify the additional cognitive load imposed by stateless versus memory-aware interactions. Hypothesis: relative to

Condition C (CAMA), Condition A (stateless) will produce higher Frustration and Effort scores on emotionally significant interactions, with divergence increasing over time.

4.2 Behavioral Proxies

System-level metrics can supplement self-report by capturing behavioral signatures of continuity burden without requiring active participant effort:

Disclosure depth over time. Mean word count per emotionally significant disclosure, or comparable semantic-depth metrics, tracked longitudinally. Hypothesis: disclosure depth decreases over time in Condition A as users learn that depth is not retained, while remaining stable or increasing in Condition C as accumulated context reduces the cost of continued disclosure.

Session initiation frequency. The rate at which users voluntarily initiate non-task-related interactions with the AI companion. Declining initiation frequency, particularly for emotional or interpersonal topics, serves as a proxy for withdrawal driven by continuity burden.

Re-disclosure ratio. The proportion of emotionally significant disclosures that repeat information previously shared in earlier sessions. In Condition A, this ratio should remain high (users must re-disclose each time). In Condition C, it should decrease as the system retains prior context. A persistently high re-disclosure ratio in Condition C would indicate retrieval failure in the memory architecture.

Latency to first personal disclosure. The time elapsed within a session before a user shares personal or emotional content. Longer latency may indicate hesitation driven by anticipation that the system will not remember prior context.

These proxies should be interpreted cautiously, as they may also be influenced by workload, fatigue, mission phase, personality, and interaction goals independent of memory architecture.

4.3 Longitudinal Analysis

The critical measurement is not any single data point but the trajectory across the study period. Continuity burden is hypothesized to be cumulative and accelerating: early sessions may show minimal difference between conditions, but divergence should increase as users in Condition A accumulate more experiences of erasure while users in Condition C accumulate more experiences of retention. Statistical analysis should employ mixed-effects models with time as a continuous predictor and condition as a between-subjects factor, testing for condition-by-time interactions that would indicate differential trajectories.

5. Safety Considerations for Persistent Memory in Spaceflight

Persistent memory introduces capabilities that stateless systems lack, but it also introduces risks that must be addressed with the same rigor applied to any mission-critical system. Table 2 summarizes the primary risks, their potential impact, the mitigation mechanisms that CAMA implements or proposes, and residual risk after mitigation.

Table 2. Risk-Mitigation Matrix for Persistent Memory in Spaceflight

Risk	Impact	Mitigation (CAMA)	Residual Risk
False-memory persistence	Incorrect inferences contaminate future interactions; crew member treated as depressed when they are not	<i>Implemented:</i> provenance tracking separates user disclosures from system inferences. Inferences are stored as provisional with explicit confidence markers. A correction mechanism downgrades false inferences without deletion, preserving the audit trail [17].	Low. Residual risk of uncorrected inferences if crew member does not engage in correction; mitigated by periodic review protocol.
Privacy and power asymmetry	System possesses sensitive personal information; potential for surveillance or misuse of intimate disclosures in ground-facing reports	<i>Implemented:</i> per-memory consent levels keep high-sensitivity memories out of default retrieval; memories can be queried on request. <i>Proposed:</i> personal-only disclosures excluded from ground-facing summaries; a pre-mission crew consent model.	Medium. Residual risk of institutional pressure to override privacy boundaries during emergencies; requires clear governance protocols.
Retrieval-induced amplification	Repeated retrieval of negative memories reinforces negative emotional states; system inadvertently deepens distress	<i>Implemented:</i> when negative affect is detected, retrieval adds coping-oriented counterweight memories. <i>Proposed:</i> therapeutic policies so negative memories are not preferentially surfaced, and decaying weight for repeatedly accessed distressing content.	Low–Medium. Therapeutic retrieval policy design requires validation in analog environments; optimal decay parameters not yet established.
Adversarial memory insertion	One crew member manipulates system memory to alter how AI interacts with another crew member	<i>Implemented:</i> every memory carries source attribution, so insertions can be traced in a provenance audit. <i>Proposed:</i> contradiction detection that flags conflicting information, and multi-user boundaries preventing	Low. Residual risk of sophisticated manipulation that mimics legitimate input; mitigated by periodic provenance audits.

		one user’s inputs from silently altering another’s model.	
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The risk-mitigation framework above is designed to be testable within the evaluation conditions proposed in Section 6. Condition D (adversarial probing) specifically targets the adversarial insertion and false-memory persistence risks, while Conditions A through C provide comparative data on privacy behavior and retrieval patterns.

5.2 Counterarguments: When Memory May Worsen Outcomes

A rigorous assessment of persistent memory must consider scenarios in which memory itself introduces harm. Several counterarguments deserve direct engagement:

Chilling effects on disclosure. If crew members know that their disclosures persist and may be included in ground-facing summaries, they may strategically under-disclose—sharing less than they would with a system they perceive as transient. This is a genuine risk. It could be mitigated by configurable disclosure boundaries proposed for CAMA (crew members could mark disclosures as personal-only), but the mitigation depends on crew members trusting those boundaries. Building that trust requires transparency about what is stored, what is shared, and under what circumstances overrides occur—a governance challenge that extends beyond architecture into institutional policy.

Over-reliance on AI companionship. A system that remembers, adapts, and demonstrates continuity may become a preferred social partner over fellow crew members, particularly during periods of interpersonal tension. This substitution could weaken crew cohesion rather than support it. The system should be designed with awareness of this risk—for example, by encouraging crew-to-crew interaction rather than positioning itself as a replacement for human connection.

Pathological framing through persistent inference. If the system stores a provisional inference (“crew member may be experiencing early depressive symptoms”) and that inference influences subsequent interactions—even subtly, through retrieval weighting or conversational tone—it may create a self-reinforcing cycle in which the system’s framing shapes the crew member’s self-perception. Provenance tracking and the provisional/durable distinction mitigate this, but the risk of confirmation bias in longitudinal AI systems remains an active area of concern that requires empirical investigation.

Ground-control power dynamics. Persistent memory summaries sent to ground-based support teams alter the information asymmetry between crew and mission control. A flight surgeon who receives a longitudinal AI-generated assessment of a crew member’s psychological state holds

information that the crew member may not know has been shared or may not agree with. This introduces command-structure implications that must be addressed through clear pre-mission agreements about summary content, frequency, crew review rights, and the conditions under which a crew member can contest a system-generated assessment.

These counterarguments do not undermine the case for persistent memory; they refine it. The claim advanced here is not that persistent memory is without risk, but that the risks of memory are manageable through architectural design and governance, while the risks of statelessness are structural and only partially mitigable, because a system that does not retain and integrate the information needed for continuity-sensitive support cannot fully compensate for that absence through interface design alone.

6. Proposed Evaluation Framework

To evaluate the hypothesis that persistent memory functions as mission-critical infrastructure, we propose a four-condition evaluation design adapted from the framework presented in Reinhold (2026) [17]:

Condition A: Baseline stateless (no persistent memory). The AI companion operates as current systems are publicly described, with no retention between sessions. This represents the current state of deployed and proposed spaceflight AI.

Condition B: Flat memory (append-only log, no provenance). The system retains a transcript of prior interactions but without emotional keying, provenance tracking, or structured retrieval. This represents a naive implementation of persistent memory.

Condition C: Provenance-aware persistent memory (CAMA). The full architecture with three-layer storage, provenance tracking, emotional keying, and correction mechanisms.

Condition D: CAMA with adversarial probing. The full architecture subjected to adversarial conditions: deliberately inserted false information, contradictory disclosures, and attempts to manipulate the system’s memory to test robustness.

Benchmark tasks should include: (1) detection of slow-onset behavioral drift across simulated 30-day timelines, as illustrated in the worked example (Section 3.3); (2) tracking and contextualizing recurring interpersonal conflicts; (3) maintaining appropriate correction behavior when user contradicts prior system inferences; (4) resisting adversarial memory insertion while preserving legitimate memory function; and (5) generating accurate longitudinal summaries for simulated

ground-control communication windows. Continuity burden should be measured across all conditions using the instruments and proxies described in Section 4.

Evaluation environments should include spaceflight analog facilities such as the NASA Human Exploration Research Analog (HERA), the Mars Desert Research Station (MDRS), or Antarctic research stations, which provide isolated, confined environments with the operational constraints relevant to assessing persistent memory in realistic conditions.

6.1 Proposed Trial Parameters

A minimum viable analog trial would consist of $N=16$ participants organized into four teams of four, with each team randomly assigned to one of the four conditions (A through D). Trial duration should be a minimum of 30 continuous days in an analog habitat, with 45–60 days preferred to capture the cumulative trajectory effects hypothesized by the continuity burden model. The Continuity Burden Inventory (CBI) should be administered weekly. The NASA TLX Frustration and Effort subscales should be administered after any interaction rated above 3 on a 5-point emotional intensity self-assessment. Behavioral proxies (disclosure depth, session frequency, re-disclosure ratio, disclosure latency) should be logged automatically by the system throughout the trial period. Mixed-effects models should test for condition-by-time interactions. Because each condition is assigned to a single team, team and condition effects are confounded in this minimal design; a fully powered trial should assign several teams to each condition.

7. Implementation Considerations

This paper presents a framework and architecture rather than a hardware specification. However, the feasibility question raised by the spaceflight context warrants brief acknowledgment. The CAMA architecture has a working open-source implementation (<https://github.com/LoriensLibrary/cama>) currently operating as an MCP (*Model Context Protocol*) server integrated with a large language model pipeline.

7.1 Empirical Performance Data

The current CAMA prototype, running on a consumer desktop system, has been used in a naturalistic single-user setting since March 2026, seeded with a chat history that begins in January 2025. As of March 2026, the system stores 52,622 memories (47,270 durable, 178 provisional, 5,174 expired) with 52,615 semantic embeddings, 879 relational edges, and a 30-slot active ring buffer. Its SQLite database, including embeddings, occupies approximately 522 MB, about 10 KB per memory.

For a six-person crew on a 30-month Mars mission, assume each crew member generates memories at twice the average rate of this corpus (52,622 memories over about 14 months, roughly 120 per day). At about 10 KB per memory, total storage would be roughly 13 GB, a modest requirement for current solid-state hardware. Retrieval latency in the current prototype, using blended semantic and keyword search against the full 52,000+ memory corpus, operates within interactive response times on consumer hardware. Adaptation for spaceflight would require radiation-hardened storage, redundant memory stores, and computational optimization for resource-constrained environments—engineering challenges that are significant but well within the domain of existing spacecraft computing design.

7.2 Correction Interface

CAMA’s correction mechanism is designed for minimal interaction friction. When the system surfaces an inference that the user recognizes as incorrect, the correction workflow is conversational: the user simply contradicts the inference in natural language (“That’s not right—I wasn’t upset about the workload, I was frustrated about the comm delay”). The system stores the correction as a new memory with provenance type “correction,” links it to the original inference, and downgrades the original’s retrieval weight without deletion. In a spaceflight implementation, this could be supplemented by a lightweight confirmation interface—a brief prompt after system-initiated references to stored memories (“I recall you mentioned difficulty sleeping. Is that still relevant? Yes / No / Clarify”)—to reduce the burden of unprompted correction while maintaining user agency over the memory model.

7.3 Group and Private Memory Partitioning

The current CAMA implementation operates as a single-user system. Adaptation for multi-crew spaceflight introduces the question of group-level events (e.g., a tense crew meeting, a shared celebration, a collective decision). The architecture supports this extension through memory provenance: group events can be stored with shared attribution while individual interpretations and emotional responses remain partitioned to each crew member’s private memory space. Detailed specification of group memory policies, including the interaction between shared and private memory layers and their representation in ground-control summaries, is deferred to Paper 2 (Permanent Extraterrestrial Habitation), where ongoing multi-user dynamics are a central concern.

8. Limitations

This paper is primarily conceptual and architectural rather than empirical. The continuity burden construct proposed here has not yet undergone psychometric validation, and the worked scenario

presented in Section 3.3 is illustrative rather than derived from analog trial data. The Continuity Burden Inventory (CBI) items described in Section 4.1 are proposed rather than validated; discriminant validity against adjacent constructs (cognitive load, trust erosion, disclosure fatigue, learned helplessness) has not yet been established empirically. Similarly, while CAMA provides a functioning open-source reference implementation with over 52,000 stored memories in naturalistic use, its suitability for spaceflight environments remains to be demonstrated through hardware qualification, human-subject evaluation in analog settings, and formal safety assessment. The behavioral proxies proposed in Section 4.2 may be confounded by factors independent of memory architecture, including workload variation, crew schedule, mission phase, novelty effects, and individual differences in communication style. Accordingly, the claims advanced here should be understood as a research and engineering agenda grounded in known operational constraints, not as proof of deployed efficacy. The evaluation framework proposed in Section 6 is designed to address these limitations through controlled analog trials.

9. Broader Implications and Future Work

This paper is the first in a four-part series examining persistent memory architecture as critical infrastructure across domains where continuity burden has measurable consequences:

Paper 2: Permanent Extraterrestrial Habitation. Extends the spaceflight analysis to permanent lunar and Martian settlements, where AI memory must span years of habitation, track generational knowledge transfer, and maintain base operational continuity across personnel rotations. This paper will address ground-to-crew communication protocols in detail.

Paper 3: Chronic Healthcare Continuity. Applies the continuity burden framework to longitudinal healthcare, where patients with chronic conditions experience the same re-disclosure fatigue with stateless systems that astronauts face—at lower intensity but over potentially longer timescales.

Paper 4: Veteran Reintegration and Trauma-Informed Care. Examines the application of persistent memory architecture to psychological support for veterans, with particular attention to music-mediated emotional continuity as a non-verbal pathway for populations who cannot or will not articulate their experiences through traditional therapeutic discourse.

Collectively, these papers argue that persistent, provenance-aware memory is not a feature enhancement for AI systems—it is foundational infrastructure for any context where the AI serves as a longitudinal partner in human well-being. The failure to implement persistent memory is not a

neutral design choice; it is a decision to accept continuity burden as an externality borne entirely by the user.

10. Conclusion

The current trajectory of AI development for long-duration spaceflight focuses on emotional intelligence, natural language processing, and diagnostic capability. These are necessary but insufficient. An AI companion that can recognize emotional distress but cannot remember that the same crew member expressed similar distress last week—that can offer conflict resolution strategies but cannot recall that the same conflict has recurred three times—that can assess symptoms but cannot track their trajectory over months—is a system that fails at precisely the moments when it matters most.

Persistent memory, implemented with provenance-awareness and appropriate safety mechanisms, transforms AI from a reactive tool into a longitudinal partner capable of supporting human psychological health across the unprecedented timescales and isolation conditions of deep space exploration. As humanity prepares to send crews to Mars, the burden of proof should shift to any proposal to deploy onboard psychological support AI without robust longitudinal memory.

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